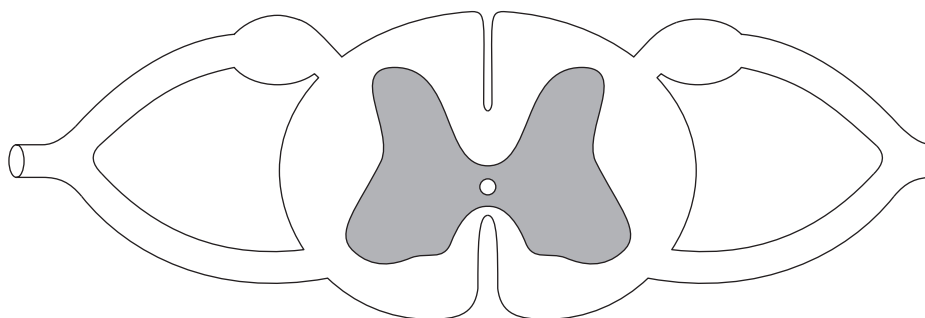


Answer all questions.

1. The diagram shows a transverse section of the spinal cord.

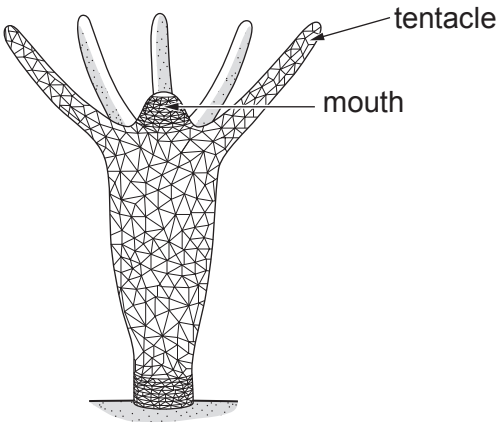


- (a) (i) **Draw and label the position** of a sensory neurone and a motor neurone and label the cell bodies. [3]
- (ii) When a person touches a hot object with their finger they will move away from the heat. State the advantage to humans of having this type of response to an external stimulus. [1]
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(b) Cnidaria is a phylum of invertebrate animals that are found in aquatic habitats. Their bodies consist of a jelly like substance sandwiched between two epithelial layers.

The diagram below shows the nervous system of a Cnidarian.



(i) State the name given to this type of nervous system. [1]

(ii) Explain why the non-myelinated neurones of Cnidarians result in slower nerve impulses than in mammals. [2]

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(b) Explain the results seen in tube 1. [3]

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(c) When the chloroplasts are isolated, there may also be mitochondria present in the sample. Explain how the results show that mitochondria are not responsible for reducing the DCPIP in this experiment. [3]

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(d) Explain why, when students increased the carbon dioxide concentration present in the sample of chloroplasts, the results of the experiment were unchanged. [2]

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